Lecture 30 Factor Analysis II: Rotation and Scores

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

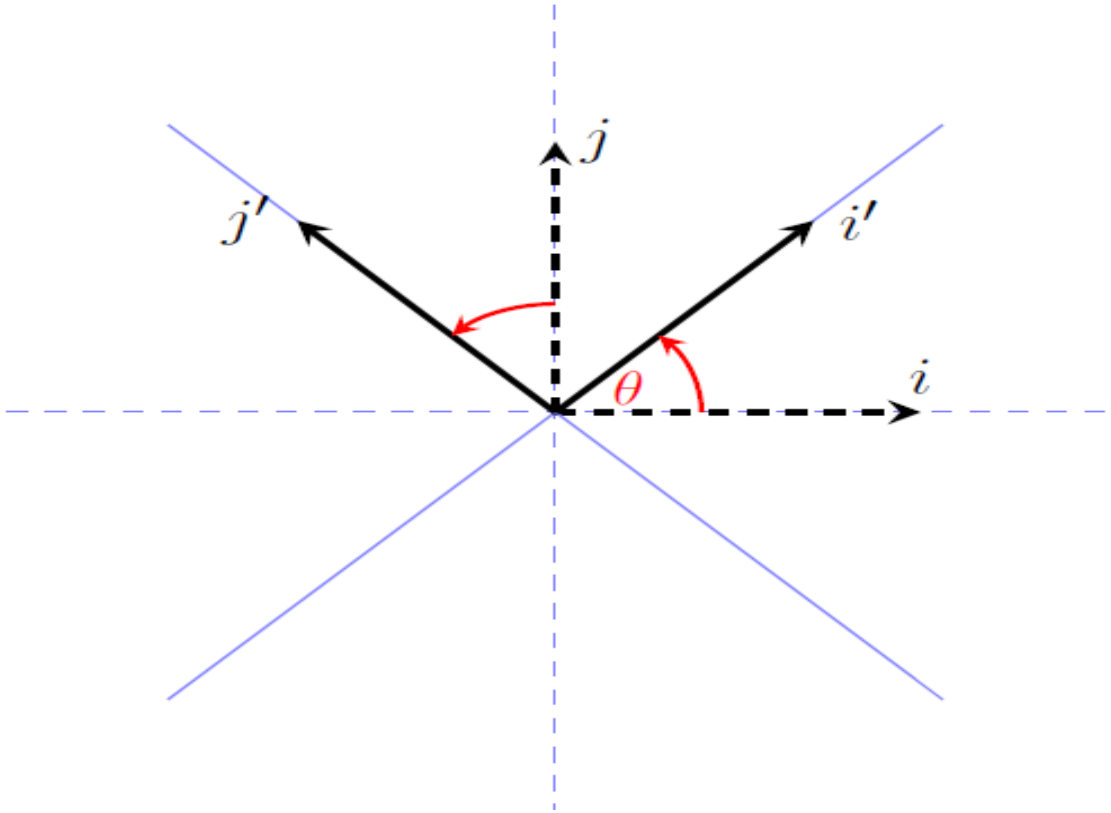
1. Rotation
2. Factor Scores
3. Other Remarks
4. Summary and Remark

Rotation

- The FA model is invariant under orthogonal transformations $\mathbf{L} \to \mathbf{L}\boldsymbol{\Gamma} = \tilde{\boldsymbol{L}}$, with $\boldsymbol{\Gamma}_{m\times m}$ orthogonal.
- One generally attempts to choose a transformation to make the factors more interpretable,
- e.g. so that the loadings on each particular factor split into interpretable groups – those with large loadings, and the others (for each $j = 1, .., m$, $\tilde{l}_{ij}^2$ should be highly varied).

Rotation Matrices

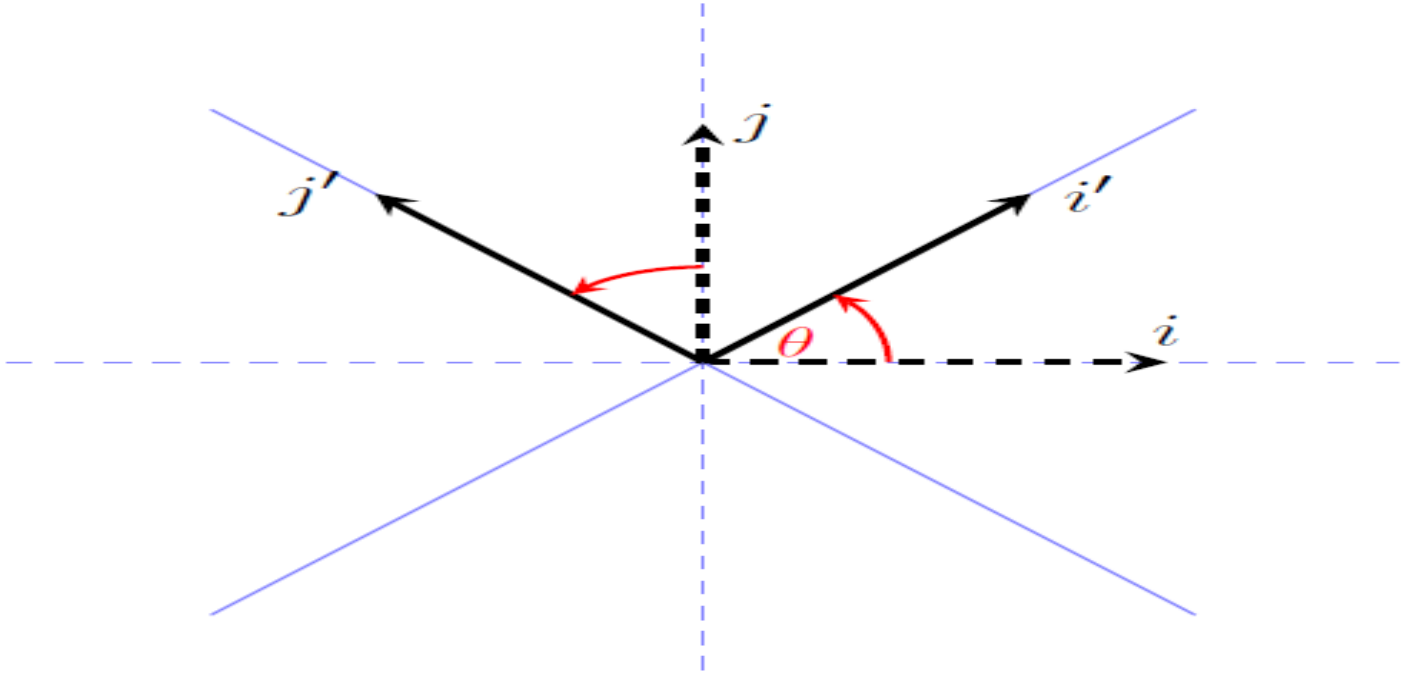
- Have a co-ordinate system in terms of $\vec{i}, \vec{j}$ orthonormal vectors
- Rotate the axes through an angle θ.



Equations of Rotation

If a point on the plane is denoted in terms of $\vec{i}$ and $\vec{j}$ by (x, y), its position in terms of the rotated basis vectors is

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta$$



In Matrix Form

The equations of rotation

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta.$$

May be written

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{\Gamma} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Using the identities $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$, rotate back through an angle of $-\theta$.

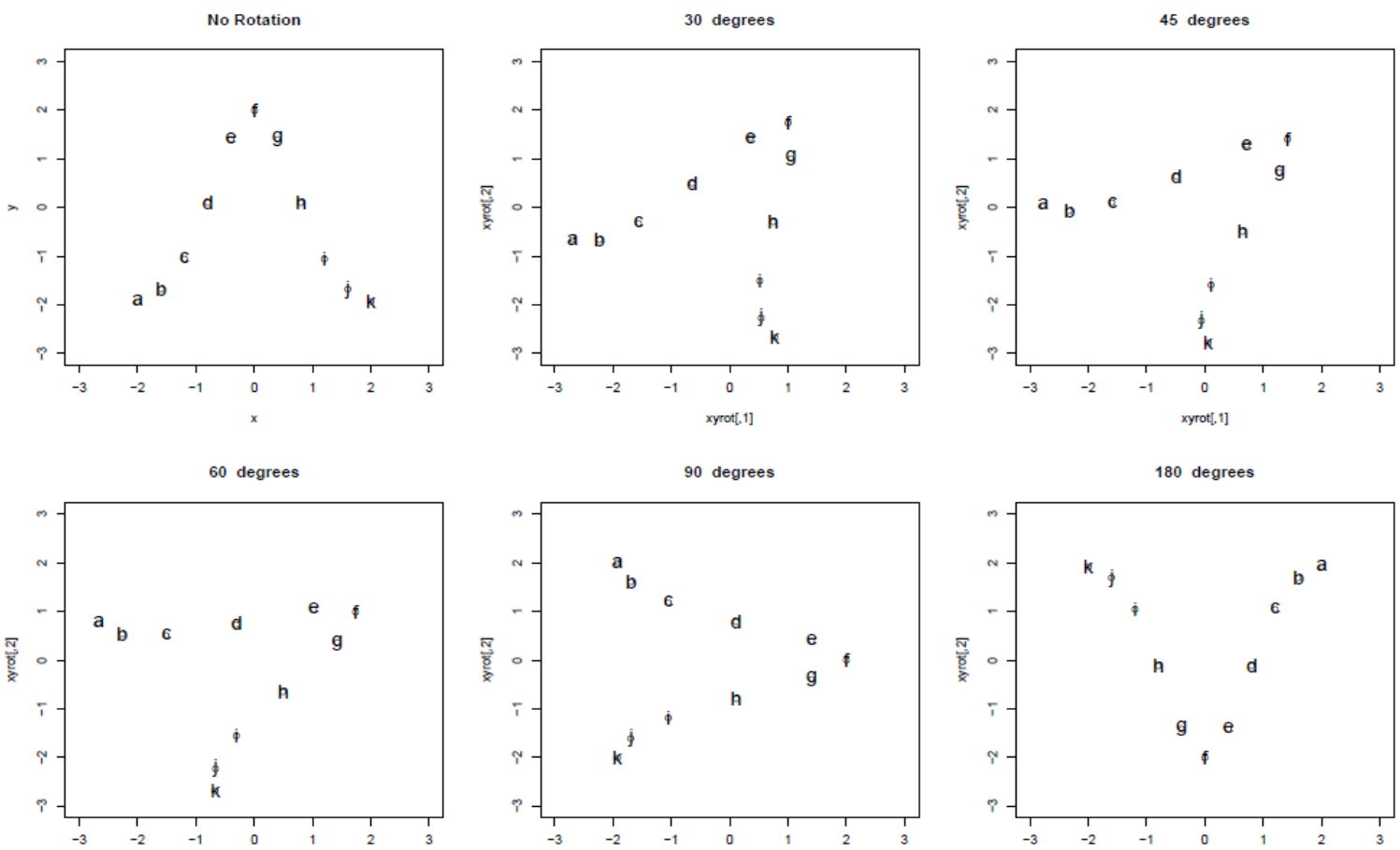
$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} = \mathbf{\Gamma}^\top \begin{pmatrix} x' \\ y' \end{pmatrix}$$

Verifying that $\mathbf{\Gamma}^\top = \mathbf{\Gamma}^{-1}$

$$
\begin{aligned}
\mathbf{\Gamma}\mathbf{\Gamma}^\top &= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\
&= \begin{pmatrix} -\cos^2\theta + \sin^2\theta & -\cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta + \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}.
\end{aligned}
$$

In higher dimension as well, pre-multiplication by an orthogonal matrix corresponds to a rotation or possibly a reflection.

Visualization of 2D Rotation



Another source of non-identifiability

$$
\begin{aligned}
\operatorname{cov}(\mathbf{x}) &= \boldsymbol{\Sigma} \\
&= \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi} \\
&= \mathbf{L}\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}\mathbf{L}^\top + \boldsymbol{\Psi} \\
&= \left(\mathbf{L}\boldsymbol{\Gamma}^\top\right)\left(\mathbf{L}\boldsymbol{\Gamma}^\top\right)^\top + \boldsymbol{\Psi} \\
&= \mathbf{L}_2\mathbf{L}_2^\top + \boldsymbol{\Psi}
\end{aligned}
$$

Infinitely many rotation matrices produce the same $\boldsymbol{\Sigma}$, even though the factor loadings in $\mathbf{L}_2 = \mathbf{L}\boldsymbol{\Gamma}^\top$ can be very different for different $\boldsymbol{\Gamma}$ matrices.

Rotating the Factors

Post-multiplication of $\mathbf{L}$ by $\mathbf{\Gamma}^\top$ is often called "rotation of the factors."

$$\begin{aligned}\mathbf{x} &= \mathbf{L}\mathbf{f} + \varepsilon \\ &= \left(\mathbf{L}\mathbf{\Gamma}^\top\right)(\mathbf{\Gamma}\mathbf{f}) + \varepsilon \\ &= \mathbf{L}_2\mathbf{f}' + \varepsilon\end{aligned}$$

- $\mathbf{f}' = \mathbf{\Gamma}\mathbf{f}$ is a set of rotated factors.
- All rotations of the factors produce the same covariance matrix of the observable data.

- Place some restrictions on the factor loadings, so that the only rotation matrix that preserves the restrictions is the identity matrix.
- Generally, the restrictions may not make sense in terms of the data. Don't worry about it.
- Estimate the loadings, perhaps by maximum likelihood.
- All (orthogonal) rotations result in the same maximum value of the likelihood function.
- Pick a rotation that results in a simple pattern in the factor loadings, one that is easy to interpret.

Simple Structure (Something like this would be nice)

$$\mathbf{\Lambda} = \begin{pmatrix} 0.87 & 0.00 \\ -0.95 & 0.03 \\ 0.79 & 0.00 \\ 0.00 & 0.88 \\ 0.01 & 0.75 \\ 0.02 & -0.94 \\ 0.00 & -0.82 \end{pmatrix}$$

Rotation to Simple Structure

- Used to be subjective, and done by hand!
- Now it's objective and done by computer.
- There are various criteria. They are all iterative, taking a number of steps to approach some criterion.
- The most popular rotation method is varimax rotation.

Varimax Rotation

- The original idea was to maximize the variability of the squared loadings in each column.
- The results weren't great, so they fixed it up, expressing each squared factor loading as a proportion of the communality.
- In practice, varimax rotation tends to maximize the squared loading of each observable variable with just one underlying factor.

Varimax criterion

- Note that
$$h_i^2 = \sum_j l_{ij}^2 = \left(\mathbf{L}\mathbf{L}^T\right)_{ii} = \left(\tilde{\boldsymbol{L}}\tilde{\boldsymbol{L}}^T\right)_{ii} = \sum_j \tilde{l}_{ij}^2;$$
define
$$v_{ij} = \tilde{l}_{ij}^2 / h_i^2, \ (\sum_j v_{ij} = 1).$$
- The varimax criterion seeks a transformation which maximizes their variances $\left\{s_j^2\right\}$:
$$\sum_{j=1}^{m} \left\{ \frac{1}{p-1} \sum_{i=1}^{p} \left(v_{ij} - \bar{v}_{\cdot j}\right)^2 \right\} = \sum_j s_j^2.$$
- `varimax(loadings(fit))`

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Weighted Least Squares Method

- We are seeking estimates of the values attained by these random factors.
- The methods to be described treat the estimates of $\mathbf{L}$ and $\boldsymbol{\Psi}$ as known; the estimates are then plugged-in.
- Bartlett scores on R, when using `factanal()`.
- If $\mathbf{L}$, $\boldsymbol{\Psi}$ and $\boldsymbol{\mu}$ are known, then

$$\mathbf{x} - \boldsymbol{\mu} = \mathbf{L}\mathbf{f} + \boldsymbol{\varepsilon},$$

  with $\boldsymbol{\varepsilon} \sim (\mathbf{0}, \boldsymbol{\Psi})$.
- The wls estimate of $\mathbf{f}$ is

$$\begin{aligned} \hat{\mathbf{f}}_{m\times 1} &= \arg\min \left(\mathbf{x} - \boldsymbol{\mu} - \mathbf{L}\mathbf{f}\right)^T \boldsymbol{\Psi}^{-1} \left(\mathbf{x} - \boldsymbol{\mu} - \mathbf{L}\mathbf{f}\right) \\ &= \left(\mathbf{L}^T \boldsymbol{\Psi}^{-1} \mathbf{L}\right)^{-1} \mathbf{L}^T \boldsymbol{\Psi}^{-1} \left(\mathbf{x} - \boldsymbol{\mu}\right). \end{aligned}$$

Weighted Least Squares Method

- This is applied as

$$\hat{\mathbf{f}}_i = \left(\hat{\mathbf{L}}^T \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{L}}\right)^{-1} \hat{\mathbf{L}}^T \hat{\mathbf{\Psi}}^{-1} \left(\mathbf{x}_i - \bar{\mathbf{x}}\right), i = 1, \ldots, n.$$

- If the loadings are rotated – $\mathbf{L} \to \mathbf{L}\mathbf{\Gamma}$ – then the scores are correspondingly transformed: $\hat{\mathbf{f}} \to \mathbf{\Gamma}^T \hat{\mathbf{f}}$ (so that $\mathbf{L}\mathbf{f}$ remains fixed).

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Weighted Least Squares Method

- The factor score coefficients are the m rows of $\left(\hat{\mathbf{L}}^T\hat{\mathbf{\Psi}}^{-1}\hat{\mathbf{L}}\right)^{-1}\hat{\mathbf{L}}^T\hat{\mathbf{\Psi}}^{-1}$ – one for each factor; the $\hat{\mathbf{f}}_i$ are the n columns of $\left[\left(\hat{\mathbf{L}}^T\hat{\mathbf{\Psi}}^{-1}\hat{\mathbf{L}}\right)^{-1}\hat{\mathbf{L}}^T\hat{\mathbf{\Psi}}^{-1}\right]\mathbf{X}_{centred}^T : m \times n.$
- When the pc solution is used, it is common to take $\hat{\mathbf{\Psi}} \propto \mathbf{I}_p$; then the $\hat{\mathbf{f}}_i$ are the first m sample pcs, normalized to have sample variance $= 1$:

$$\hat{\mathbf{f}}_i = \begin{pmatrix} \frac{\hat{\gamma}_1^T(\mathbf{x}_i - \bar{\mathbf{x}})}{\sqrt{\hat{\lambda}_1}} \\ \vdots \\ \frac{\hat{\gamma}_m^T(\mathbf{x}_i - \bar{\mathbf{x}})}{\sqrt{\hat{\lambda}_m}} \end{pmatrix}.$$

Regression Method

- (regression scores on R). If the distributions are Normal, then $\mathbf{x} - \boldsymbol{\mu}$ and $\mathbf{f}$ are jointly normal:

$$\begin{pmatrix} \mathbf{x} - \boldsymbol{\mu} \\ \mathbf{f} \end{pmatrix} \sim N_{p+m}\left(\mathbf{0}, \begin{pmatrix} \boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T + \boldsymbol{\Psi} & \mathbf{L} \\ \mathbf{L}^T & \mathbf{I}_m \end{pmatrix}\right).$$

- The conditional distribution of $\mathbf{f}$, given $\mathbf{x}$, is

$$N_m\left(\mathbf{L}^T\boldsymbol{\Sigma}^{-1}\left(\mathbf{x} - \boldsymbol{\mu}\right), \mathbf{I}_m - \mathbf{L}^T\boldsymbol{\Sigma}^{-1}\mathbf{L}\right),$$

- and $\mathbf{f}$ is estimated by the conditional mean:

$$\hat{\mathbf{f}}_i = \hat{\mathbf{L}}^T\hat{\boldsymbol{\Sigma}}^{-1}\left(\mathbf{x}_i - \bar{\mathbf{x}}\right).$$

Regression Method

- One approach here – aimed at robustness against an incorrect choice of m – uses $\hat{\mathbf{\Sigma}} = \mathbf{S}$, the unconstrained sample covariance matrix:

$$\hat{\mathbf{f}}_i = \hat{\mathbf{L}}^T \mathbf{S}^{-1} \left(\mathbf{x}_i - \bar{\mathbf{x}} \right).$$

- Another proceeds by using the matrix identity

$$\begin{aligned} \mathbf{L}^T \mathbf{\Sigma}^{-1} &= \mathbf{L}^T \left(\mathbf{L}\mathbf{L}^T + \mathbf{\Psi} \right)^{-1} \\ &= \left(\mathbf{I}_m + \mathbf{L}^T \mathbf{\Psi}^{-1} \mathbf{L} \right)^{-1} \mathbf{L}^T \mathbf{\Psi}^{-1}, \end{aligned} \tag{2.1}$$

(assigned) obtaining

$$\hat{\mathbf{f}}_i = \left(\mathbf{I}_m + \hat{\mathbf{L}}^T \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{L}} \right)^{-1} \hat{\mathbf{L}}^T \hat{\mathbf{\Psi}}^{-1} \left(\mathbf{x}_i - \bar{\mathbf{x}} \right).$$

Exploratory Factor Analysis

- Factor analysis is to uncover the underlying structure of a relatively large set of variables.
- Exploratory Factor Analysis (EFA) is to identify the underlying relationships between measured variables.
- It is commonly used by researchers when developing a scale (a scale is a collection of questions used to measure a particular research topic) and serves to identify a set of latent constructs underlying a battery of measured variables.
- When no a priori hypothesis about factors or patterns of measured variables.

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Confirmatory Factor Analysis

- Confirmatory Factor Analysis (CFA) is a special form of factor analysis, most commonly used in social research.
- It is to test whether measures of a construct are consistent with a researcher's understanding of the nature of that construct (or factor).
- The objective is to test whether the data fit a hypothesized measurement model.
- This hypothesized model is based on theory and/or previous analytic research.
- It is part of broader concept, Structure Equation Model (SEM), and can be in R with package `SEM`.

FA and PCA

- In factor analysis, the researcher makes the assumption that an underlying causal model exists.
- PCA is simply a variable reduction technique and can be used to do factor analysis.
- In terms of the correlation matrix, this corresponds with focusing on explaining the off-diagonal terms (i.e. shared co-variance), while PCA focuses on explaining the terms that sit on the diagonal.
- However, as a side result, when trying to reproduce the on-diagonal terms, PCA also tends to fit relatively well the off-diagonal correlations.
- For more comparison, see Fabrigar et al. (1999). *Evaluating the use of exploratory factor analysis in psychological research*. Psychological Methods.

Summary and Remark

- Rotation
- Factor Scores